

Tyler Nielsen

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SUMMARY

Computer Systems Engineering senior at ASU with hands-on experience in embedded systems, aerospace-focused projects, and digital circuit design. Skilled in C/C++, microcontroller development, and real-time operating systems, with project work integrating electronics, communication protocols, and low-power radio. Eager to apply engineering expertise and problem-solving skills to advancing aerospace and embedded technologies.

EDUCATION

Bachelor of Science in Engineering, Computer Systems Engineering Expected May 2026
Arizona State University, Tempe, AZ 4.0 / 4.0 GPA
Relevant Coursework: Embedded Microprocessor Systems, Distributed Software Development, Operating Systems, Design of Digital Systems, Computer Architecture, Foundations of Machine Learning

SKILLS

Technical: Embedded Systems, Linux, Git/GitHub, Jira, FreeRTOS, Pico C/C++ SDK, Vivado, ARM Cortex, KiCad, Tkinter, Jupyter Notebooks

Languages: C, C++, Python, Java, Verilog, MIPS Assembly

Organizational: Project Management, Peer Tutoring, Technical Writing

PROFESSIONAL EXPERIENCE

Arizona Space Grant Consortium: NASA Space Grant Intern August 2025 – Present

- Working to build a modular space software and hardware stack to streamline development of space and high-altitude oriented projects for the Interplanetary Initiative at Arizona State University.

Tutor.com: Intermediate Tutor February 2024 – Present

- Tutoring high school and undergraduate students in math and computer science concepts, including Calculus, Linear Algebra, C, C++, Java, and Python.

Arizona State University: Undergraduate Teaching Assistant August 2025 – December 2025

- Working as part of the staff team for CSE 320 Fundamentals of Digital Design. Holding office hours and assisting students in FPGA development with Verilog using Vivado Design Suite.

Brigham Young University: Undergraduate Research Assistant April 2023 – July 2023

- Implemented Python scripts for satellite footprint simulations for the GLOWS satellite from BYU's Microwave Earth Remote Sensing Lab using custom Python libraries.

PROJECTS

Sun Devil Satellite Laboratory Coconut CubeSat Spring 2024 – Present

Software Team Lead - Led a team of 7 to create embedded flight software on an ARM Cortex-based microcontroller for a store-and-forward satellite using FreeRTOS and the Pico C/C++ SDK in partnership with NASA's CubeSat Launch Initiative:

- Integrated flight software with electronics system using multicore debugging and electronics analysis tools.
- Developed a LoRa radio task, packet definitions, command responses, and I2C device drivers.
- Created Python scripts in OpenC3 COSMOS for ground station automation, configuration, and Day-In-The-Life testing.

ASU ASCEND High-Altitude Balloon Spring 2024 – Present

Software Team Lead - Led a team of 10 to develop embedded, dual-core, modular flight software in C++ alongside multithreaded ground software in Python to provide access to real-time telemetry data for a high-altitude balloon payload:

- Designed a custom packet system, reducing needed onboard storage and telemetry data rate by 67%.
- Developed a ground station GUI with Tkinter for packet decoding and debugging with multicore serial output sorting.
- Utilized Taiga and GitHub for collaboration, progress tracking, and maintenance automation with GitHub Actions.
- Presented project conclusions and development decisions at the Arizona Space Grant Consortium.

BYU Spacecraft Club PocketQube High-Altitude Balloon Spring 2023

Project Lead - Organized a team of 10 undergraduates to build a first physical iteration for a PocketQube micro satellite and flew it on a high-altitude balloon.

NASA RASC-AL Competition Fall 2021 – Summer 2023

Sub Team Lead – Led a sub section team to research and write a Project Proposal paper, poster, and video for the Revolutionary Aerospace Systems Concepts Academic Linkage (RASC-AL) Competition:

- Proposal Themes: Suitport Logistics Carrier (2022), Homesteading Mars (2023)